

EasyCOD on-line ANALYZER

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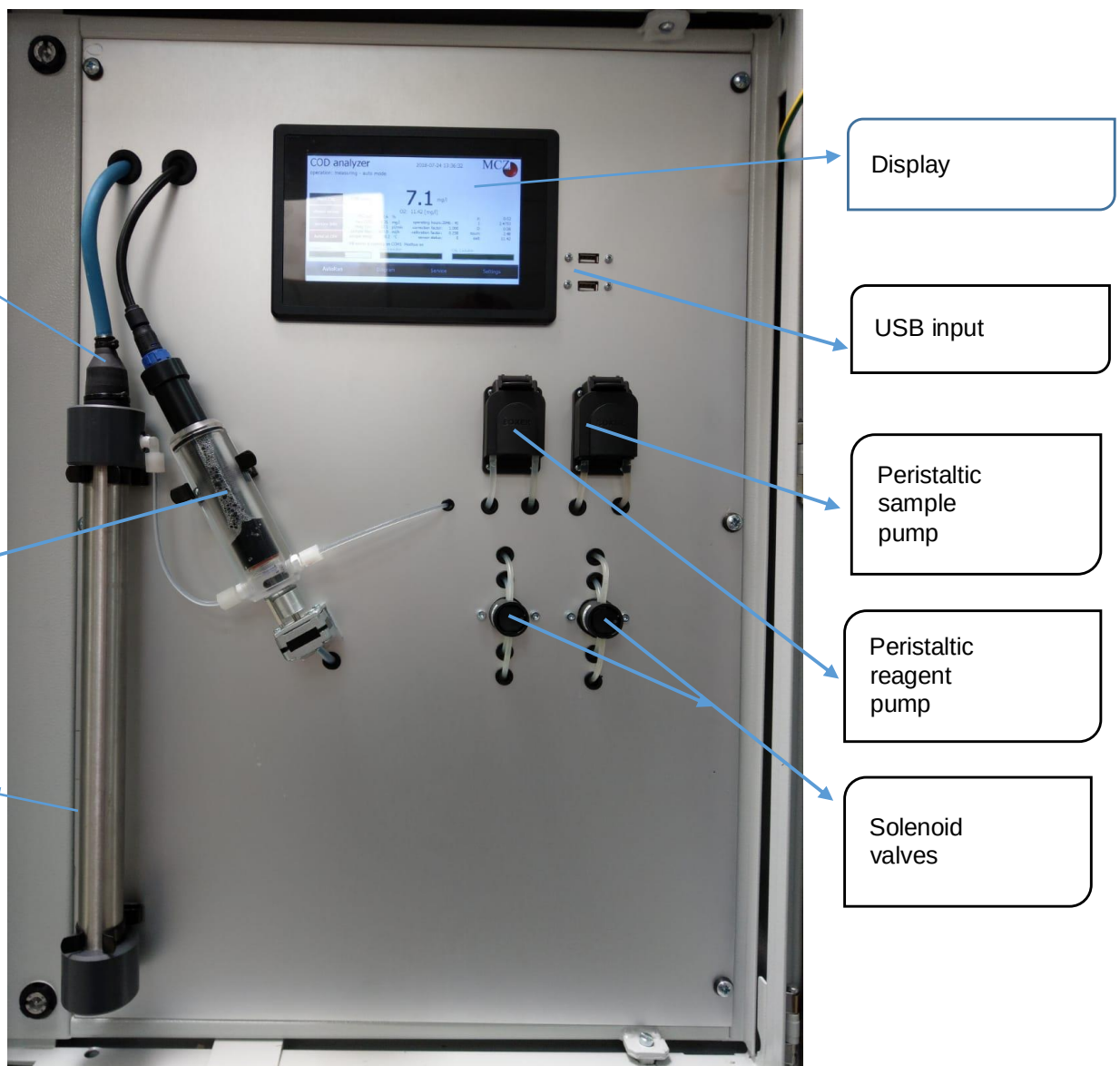
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1 Introduction

The COD (Chemical Oxygen Demand) value is the quantity of oxygen which is necessary in order to oxidize chemically all the oxidizable compounds of a water sample. The oxygen demand is therefore specified in mg O₂/l.

The EasyCOD analyzer from MCZ GmbH Umweltmesstechnik determines the COD value in an on-line process. The measured value is continuously available without delay.

The construction of the EasyCOD analyzer is shown in figure 1.



2 General

a. Safety Instruction

The EasyCOD analyzer from MCZ GmbH Umweltmesstechnik are on-line analysis instruments for determining the COD value of aqueous samples in water treatment technology. The measurement is carried out directly or with the diluted sample.

Any other use or a use going beyond this counts is improper. The operator may operate and maintain the instrument only taking into account all the advice for use in the present operation manual, and after instruction has been given by service personnel authorized by the manufacturer. The operator is obliged to comply absolutely with the prescribed maintenance operations and the associated exchange of components subject to wear.

Applications going beyond the envisaged use are inadmissible. MCZ GmbH Umweltmesstechnik accept no liability for damages which are caused by improper use of the instrument.

Proper use also includes:

- the observance of all the instructions in the operation manual,
- the observance of maintenance operations,
- carrying out inspections by service personnel,
- the observance of the general and specific safety advice in this operation manual, as well as the relevant regulations for preventing accidents,
- the observance of the usual care in working with chemicals.

b. Observe instructions in the operation manual

Safe handling and satisfactory operation of the EasyCOD online analyzer are dependent on being acquainted with all the fundamental safety regulations. The operation manual, in particular the safety instructions, are to be observed by all those persons who operate the instrument and produce calibration solutions and/or reagents. In addition, it is necessary to observe the rules and regulations for accident prevention which apply at the location of use.

The operator checks the safety-aware work of the personnel at regular intervals.

c. Obligations of the personnel

Persons who are charged with working on the EasyCOD commit themselves, before starting work:

- to observe the regulations about health and safety at work and accident prevention,
 - to read the safety Chapter and the warning instructions in this operation manual and to observe these continuously during installation, commissioning and operation.

d. Hazards in dealing with the instrument and chemicals

The EasyCOD instrument have been constructed in accordance with recognized safety rules. The instrument is to be used only - for the proper application - in a satisfactory safety state.

In the case of improper use, hazards may arise to the health of the user or third parties (severe damage to the skin or to the eyes) and/or impairments to the instrument, or to other material assets.

MCZ GmbH Umweltmesstechnik accepts no liability for damages which arise from improper dealings with the instrument or chemicals.

Faults which can impair safety should be eliminated without delay.

e. Warranty and Liability

The "general business conditions" of MCZ GmbH Umweltmesstechnik apply in principle. These are available to the operator at the latest from the time of concluding the contract.

Guarantee and liability claims are ruled out if they can be traced back to one or more of the following causes:

- improper use of the instrument,
- improper installation, commissioning, operation and maintenance of the instrument,
- operation of the instrument in the event of defective and/or non-functioning safety and protective equipment,
- non-observance of the instructions in the operation manual relating to installation, commissioning, operation and maintenance of the instrument,
- unauthorized constructional changes to the instrument,
- deficient monitoring of instrument parts which are subject to wear,
- repairs carried out improperly,
- catastrophes resulting from the action of foreign bodies and Acts of God.

f. Informal safety measures

In addition to the operation manual, the generally applicable and the local regulations for accident prevention and for environmental protection are to be provided and observed.

Check the following several times a week:

- all visible instrument components for detectable external damage,
 - all electric feed lines for loose connections, scorch points and other damage,

g. Maintenance and overhaul, rectification of faults

It is necessary to carry out the maintenance operations described in this operation manual, so that low-fault operation is ensured in the long term. If fault occur during operation with the EasyCOD then information should be sought in this operation manual. If a fault occurs in your instrument, and it cannot be rectified using the information in this operation manual, then please contact the Service Department of MCZ GmbH Umweltmesstechnik.

No changes, additions or rebuilding on the EasyCOD may be undertaken without authorization from MCZ GmbH Umweltmesstechnik.

Instrument part in a unsatisfactory condition should be exchanged immediately. Only original replacement or original wearing parts may be used. In the case of parts procured elsewhere, it is not guaranteed that they have been constructed and prepared to comply with the Operation Manual. For guarantee claims, repair and spare-parts service please contact:

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3. Technical Data

Type		EasyCOD	
Measuring range		0~200 mg/l COD	
Measuring cycle		Continuous	
Signal output	- TCPIP	HB protocol	
	- RS232	HB protocol	
Alarm output		Software stati HB	
Max. load		350 ▢	
Dilution		Without	With
Measurement accuracy		± 1 % of full scale (based on standard solution under laboratory conditions)	± 1 % of full scale x dilution factor (based on standard solution under laboratory conditions)
Repeatability		± 1 % of full scale	
Calibration	- start	Automatically or manually	
	- interval	1 – 7 days	
Interface		RS 232	
Temperature:	- medium	+ 15°C ~ + 40° C	
	- ambient	+ 10°C ~ + 35° C	
Required flow from filter		400—600 ml/h	
Sample feed		External feeding pump	
Housing		Frame construction	
Dimensions (WxHxD)		600 mm x 800 mm x 400 mm (W x H x D)	
Weight		Approx. 60 kg (without reagents)	
Type of protection		IP 54	
Connection cross section		Max. 2.5 mm ² stranded wire	
Voltage supply		230 V AC ± 10%, 50 Hz	
Power consumption		Max. 300 VA	
Preparation of the sample		Internal five sieve 0,5 mm	

4 .Function

4.1 Principle of measurement

The oxidizable components found in the sample water are oxidized by means of hydrogen peroxide (H_2O_2) as oxygen donor under the catalytic effect of UV light. H_2O_2 under UV light irradiation becomes very actively oxidizing OH radicals (OH).
UV

(1) $\text{H}_2\text{O}_2 \xrightarrow{\text{UV}} 2 \text{OH}$

These OH radicals oxidize the sample compounds to $\text{CO}_2 + \text{H}_2\text{O}$ (water) and other compounds.

(2) sample compounds + OH $\rightarrow$ $\text{CO}_2 + \text{H}_2\text{O} + \text{others}$

Remaining OH radicals which do not react with the sample water become oxygen (O_2) + water (H_2O).

(3) $4 \text{OH} \rightarrow \text{O}_2 + 2 \text{H}_2\text{O}$

All chemical reactions take place in the UV reactor and all measurements are fully continuously.

The H_2O_2 reagent is added to the sample by the means of a peristaltic pump having a variable speed. The H_2O_2 reagent is added to the sample so as the value of the DO probe behind the UV reactor is 11 mg/l O_2 . If the COD value increases the value of the DO probe decreases and vs.

In this case of COD values and therefore decreasing O_2 values the amount of H_2O_2 is automatically increased by increase of the reagent pump speed and vs. The transient effect depends on the settings of "proportional factor" for regulating the speed of the reagent pump.

4.2 Chemicals receipt

For the normal work EasyCOD is using reagent solution that can be 0,6% or 1,2% depending of how much there is COD in measured water. MCZ recommend to use 0,6% H₂O₂ in water that have 0-50mg/L COD and 1,2% above 50mg/L of COD

Reagent EasyCOD

6000 ml	3000 ml	Units	Component
100	50	ml	Concentrated sulfuric acid 30% H ₂ SO ₄ or 20ml 96%
120	60	ml	Hydrogen peroxide 30% H ₂ O ₂
5780	2890	ml	Distilled water H ₂ O

CAL 1

Cal1	Distilled water H ₂ O
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CAL 2 (100 mg/L O₂)

6000 ml	3000 ml	Units	Components
0,528	0,264	g	PHP Potassium Hydrogen Phatalate C ₈ H ₅ HO ₄
10	5	ml	Concentrated sulfuric acid 30% H ₂ SO ₄ or 2ml 96%
6000	3000	ml	Distilled water H ₂ O

4.3 Installation and Connection

Electrical connection

The electrical connection must only be performed by suitably qualified personnel.

Cable leads not in use must be sealed off in order to avoid damage from penetrating moisture.

Following complete installation, the main switch on the right side of the analyzer should be switched on.

For operation, the power supply must be ensured all the times. For functioning, the main switch must be in the "1" position

Tubings

After positioning EasyCOD inside container or similar selected location tubing must be connected according manual.

EasyCOD requires one input and one output tube. On input line tube is also fine small filter that depending on water, requires cleaning every 10-15 days.

Output tube can be connected to any output from container / location of installation.

When transporting the device and during assembly, the device must be secured in such a manner that tipping is excluded.

During transport and during erection of the EasyCOD, the instrument should be safeguarded such that tilting is ruled out.

Commissioning

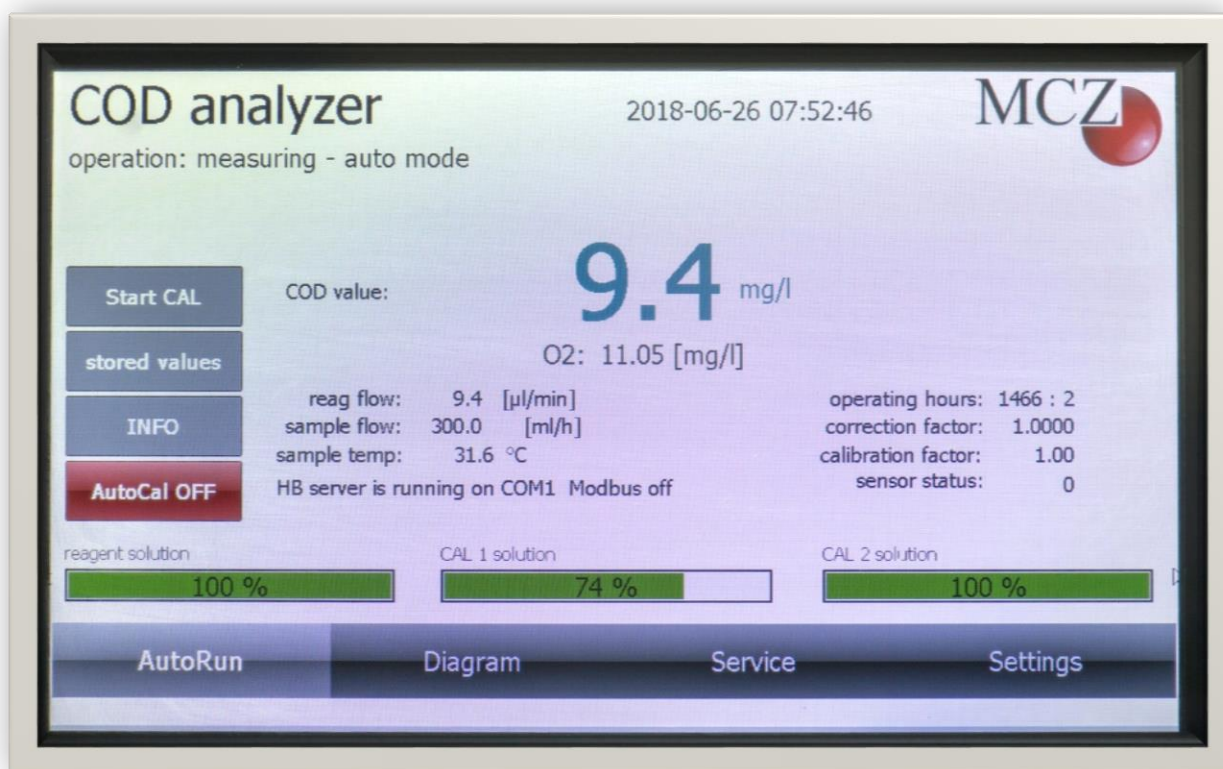
- Carry out installation
- Prepare solutions and place filled containers into the analyzer
- Switch on main switch
- Check settings
- Automatically measurement
- Comparison with laboratory measurement
- OK

4.4 Settings and Operation

The parameters were set by the manufacturer during the final check on the analyzer and must therefore not be changed.

After switching main switch on to position 1 analyzer will start automatically. Main screen will start (figure 1)

Figure 1.



On main screen we can see 4 function:

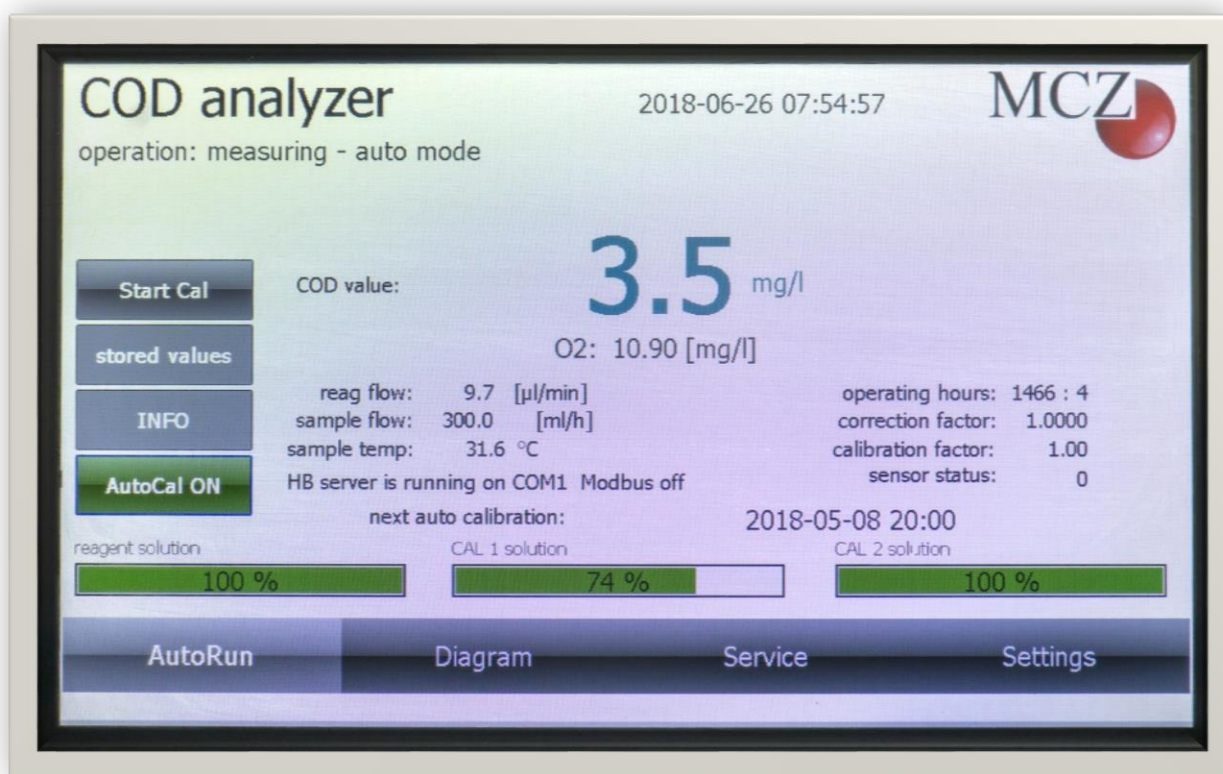
1. AutoRun
2. Diagram
3. Service
4. Settings

AutoRun:

During this function analyzer is in measuring mode (auto mode) and all data are being logged. Main screen have also sub functions like:

1. Manual Calibration
2. Stored values
3. INFO
4. AutoCal OFF/ON (figure 1 and 2)

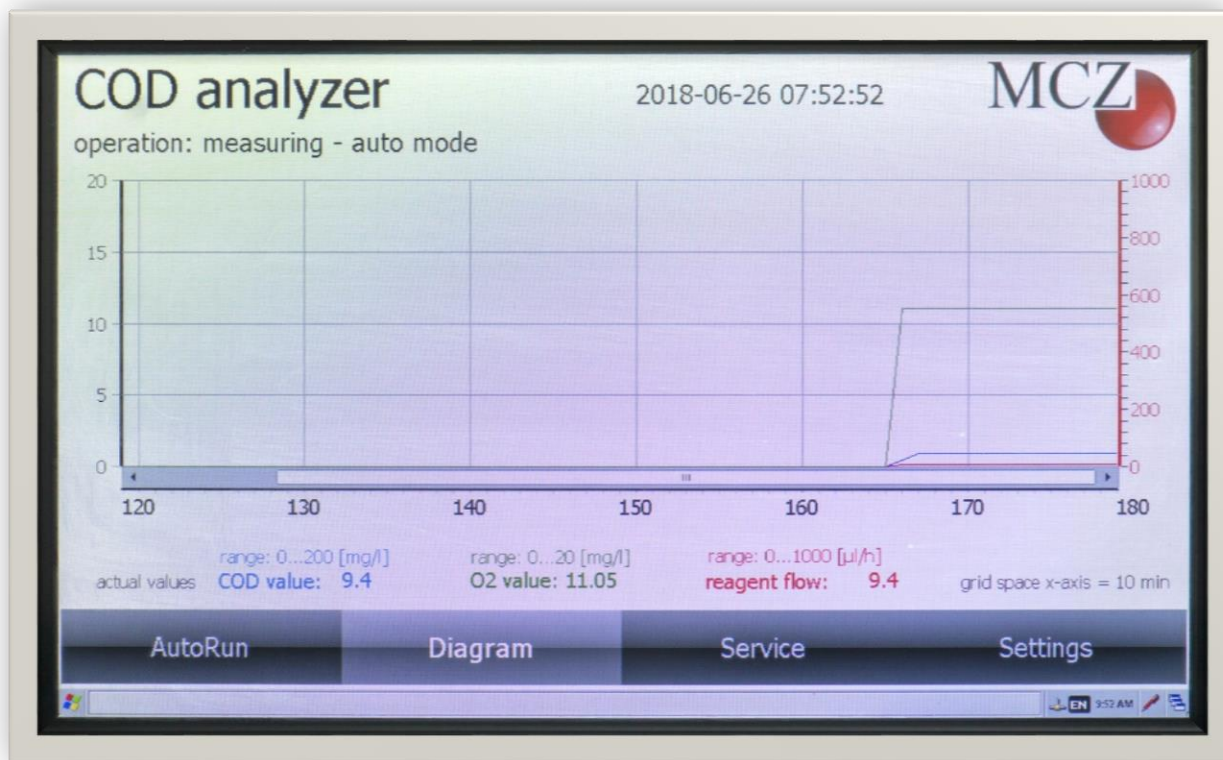
Figure 2



Diagram

In sub menu Diagram analyzer will still be on measuring mode (auto mode) and all data will be logged. In this menu we can see last 180 min of data in diagram. Grid space is 10 min (figure 3)

Figure 3.



Service

In the service menu (figure 4) analyzer is going to stand by mode and both pumps and UV lamp are shut down. Analyzer will be in stand by mode till we change menu to AutoRun or Diagram.

In this menu we can find more sub menu like:

1. Refill reagent
2. Refill CAL 1
3. Refill CAL 2
4. Cal log table
5. Cal log diagram
6. Status/error code table
7. Purge
8. Close modbus
9. COD application shut down

Figure 4.

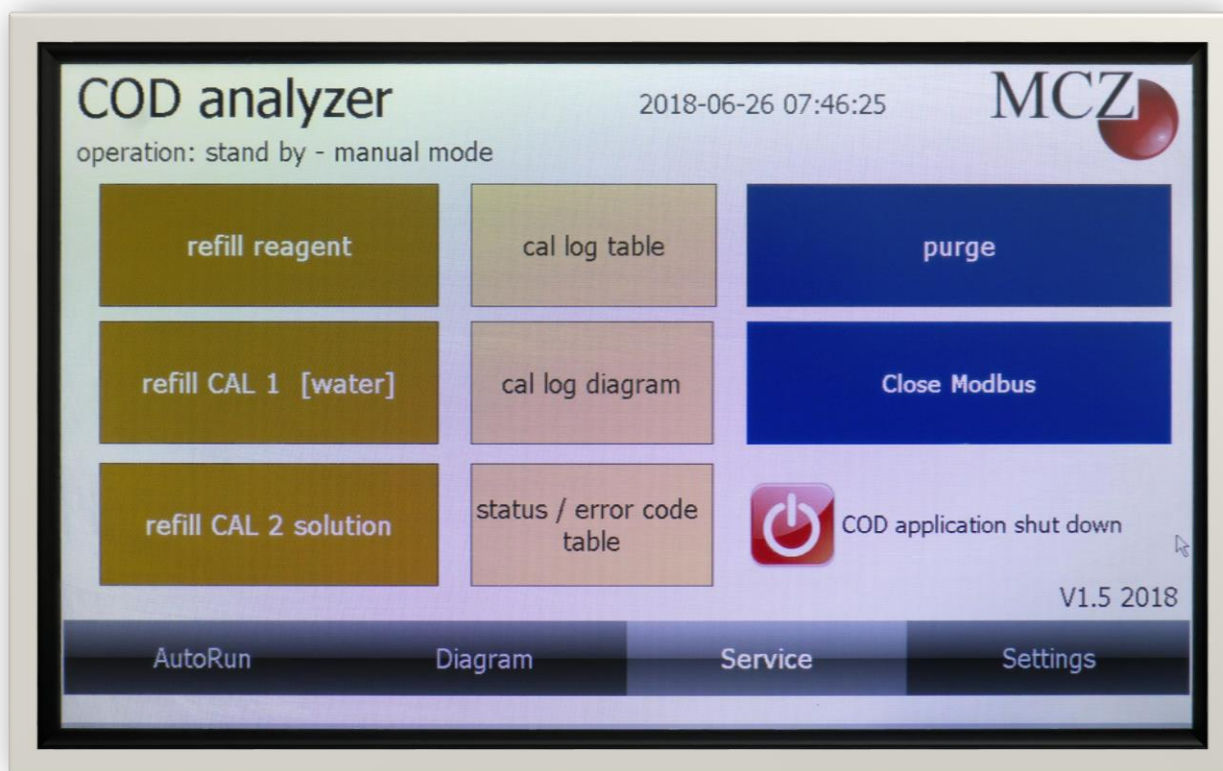
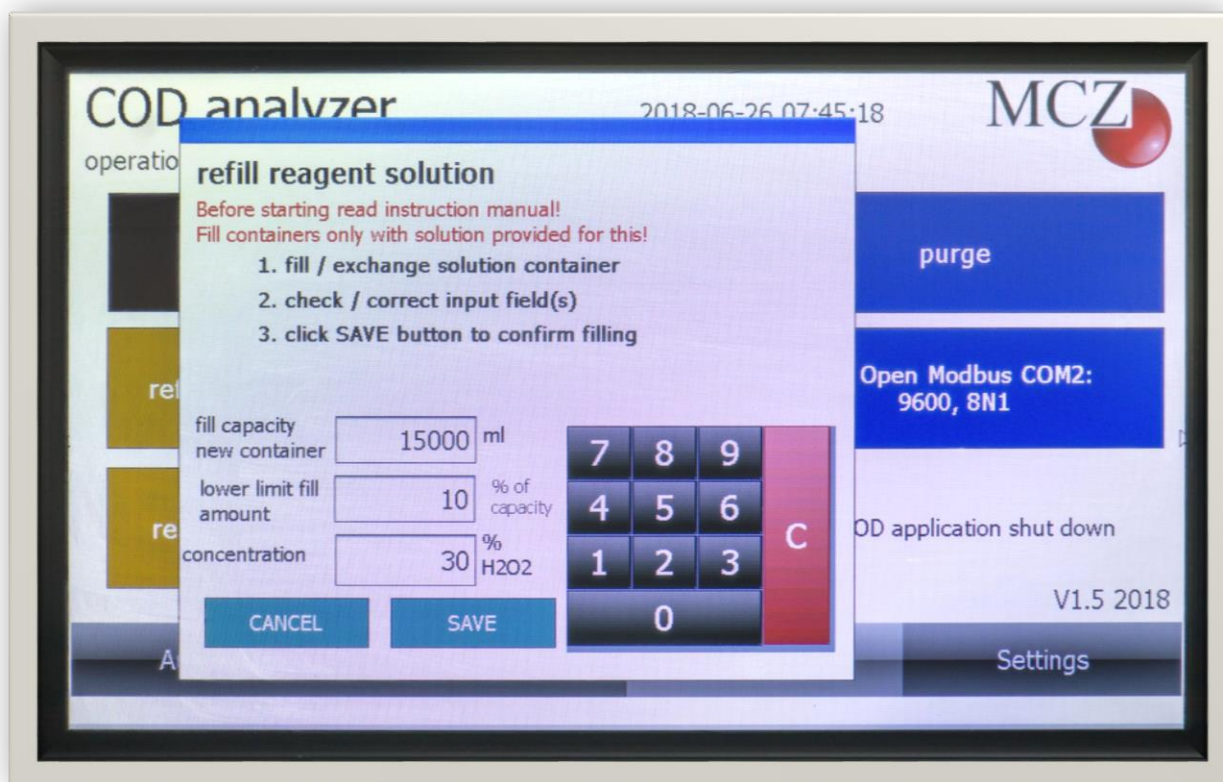


Figure 5



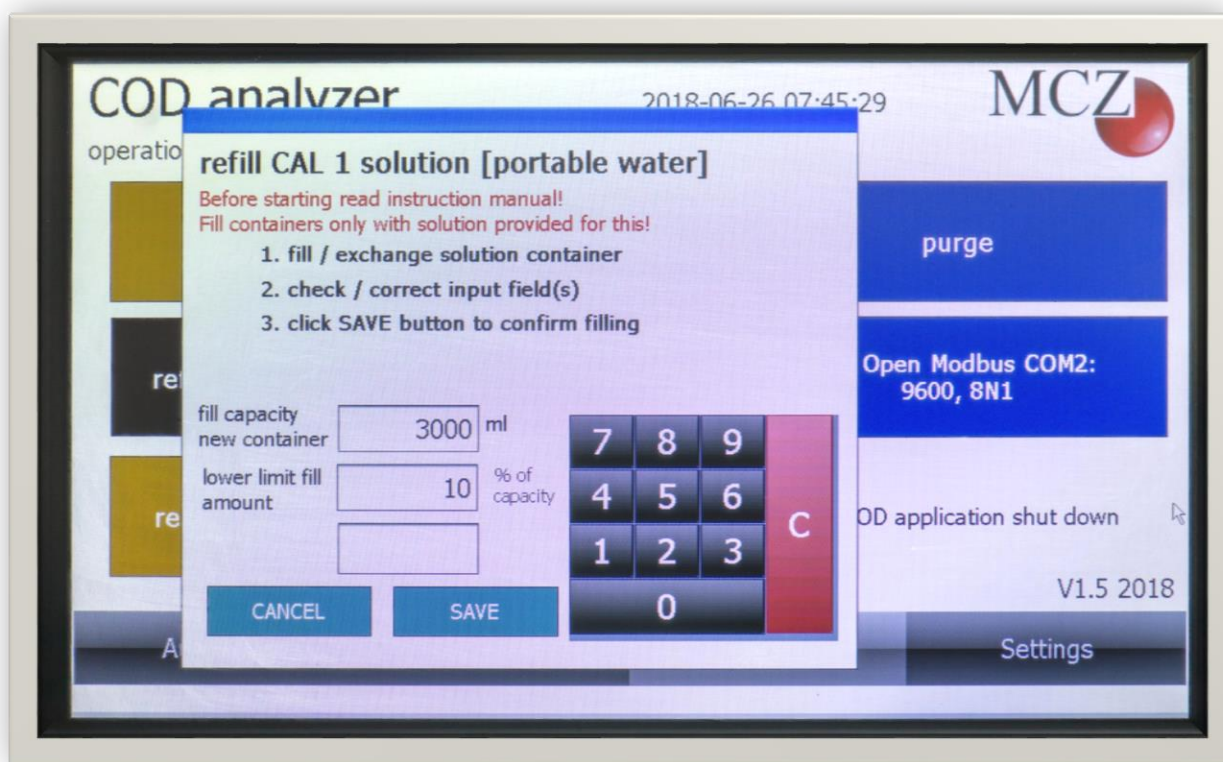
Refill reagent

In “refill reagent solution” window we can set up our reagent by adding capacity of the container, lower limit and concentration of H₂O₂.

Default setting are 6000 ml container, lower limit is 5% and H₂O₂ concentration is 30%. When we run out of reagent and/or it come to lower limit analyzer will automatically stop and go on stand by till we refill reagent and make setting in refill reagent.

Refill CAL 1

Figure 6

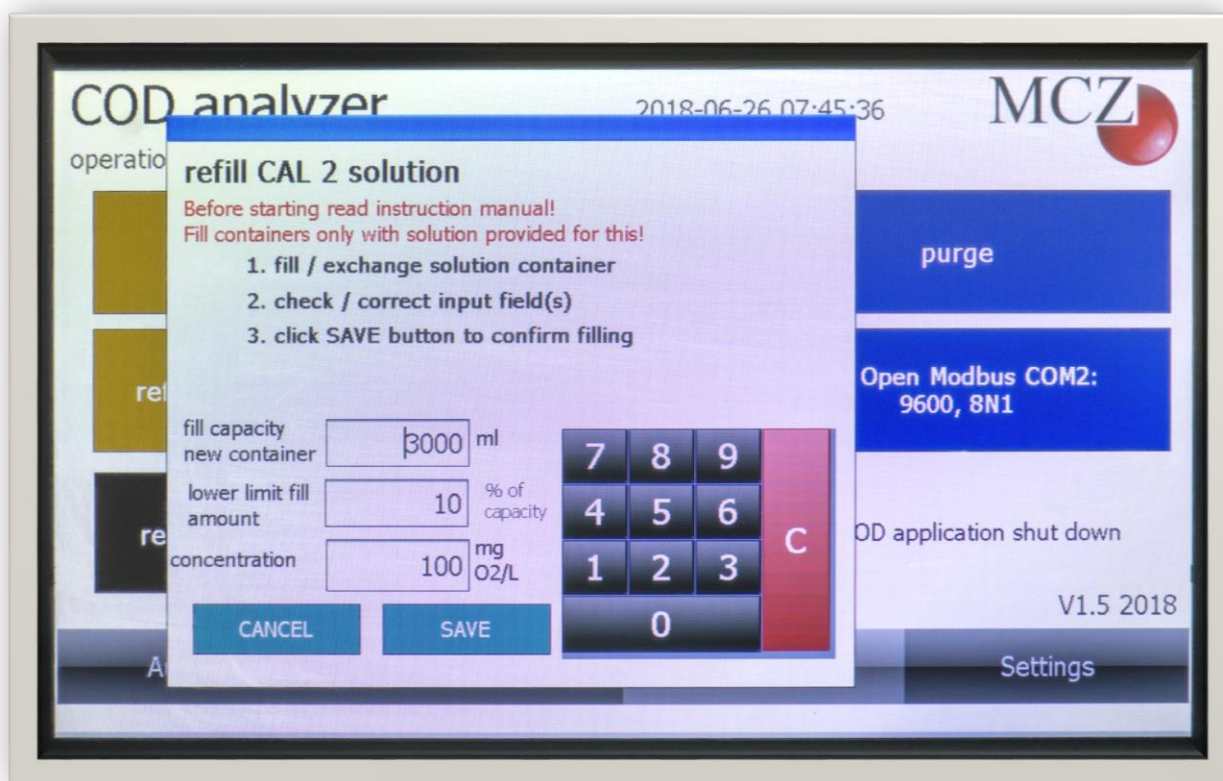


In “refill CAL 1” window we can set up our CAL 1 by adding capacity of the container and lower limit.

Default setting is 3000 ml container and lower limit is 5%. When we run out of CAL and/or it come to lower limit analyzer will automatically stop and go on stand by till we refill CAL 1 and make setting in refill CAL 1

Refill CAL 2

Figure 7



In "refill CAL 2" window we can set up our CAL 2 by adding capacity of the container, lower limit and actual concentration of COD mg/L.

Default setting is 3000 ml container, lower limit is 5%. 100 mg/L COD. When we run out of CAL and/or it come to lower limit analyzer will automatically stop and go on stand by till we refill CAL 2 and make setting in refill CAL 2.

The concentration of CAL 2 can be set by desire, MCZ recommend using 50mg/L or 100mg/L.

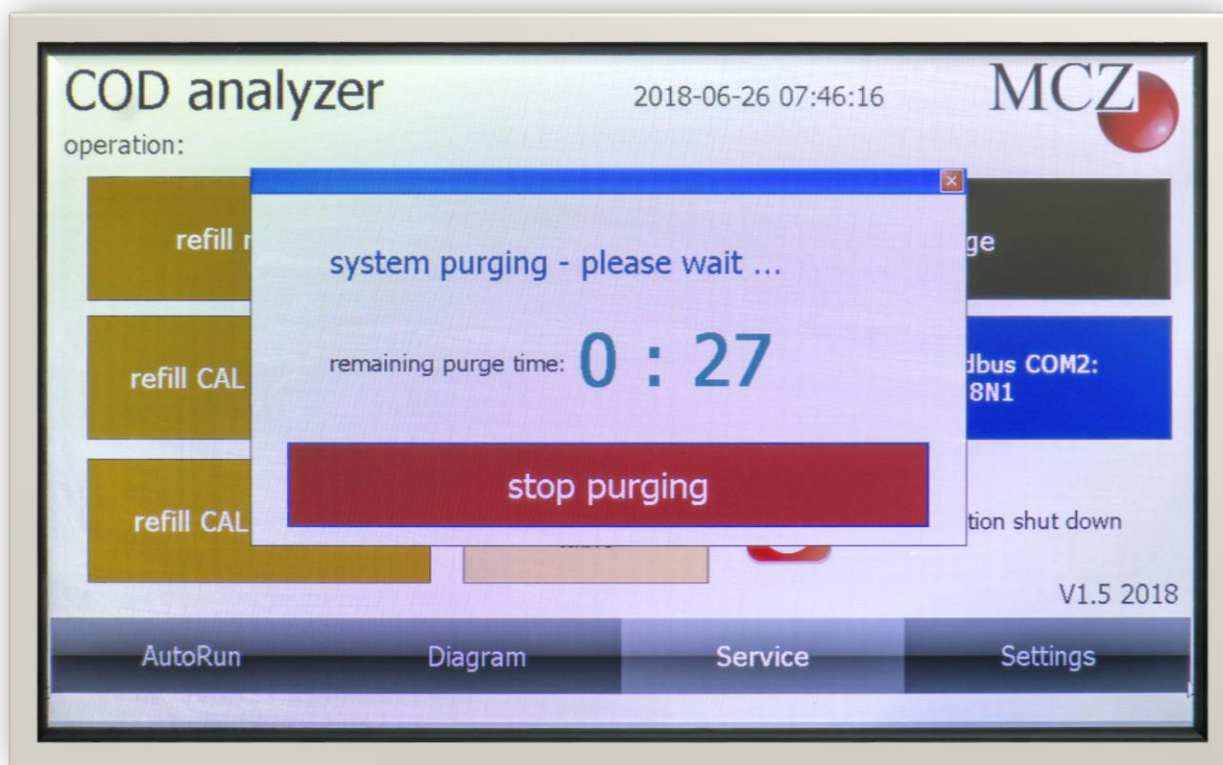
Figure 9.



Purge

EasyCOD have purge function to clean the system where both pumps working on maximum speed. Purge last 30 seconds and can not be stopped. After purge analyzer should be purged from any COD value. Figure 10

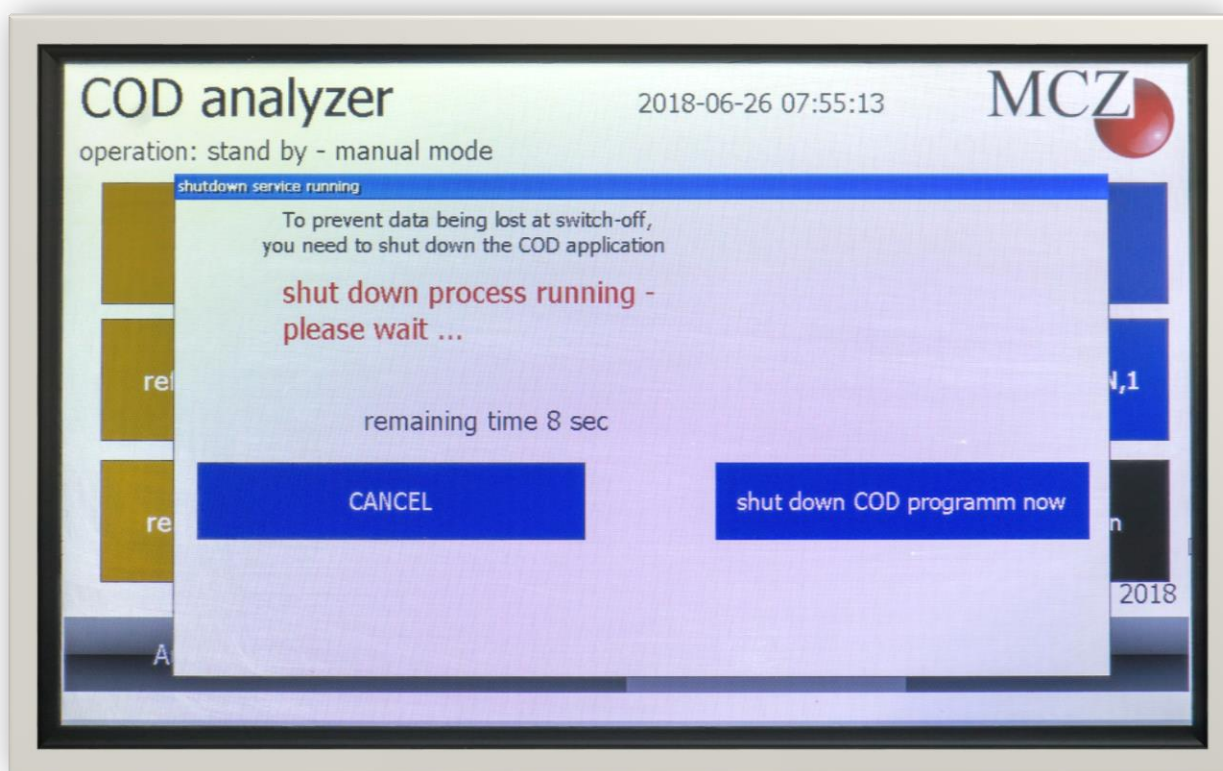
Figure 10



Shut down

If there is need for EasyCOD application to update, etc. MCZ recommend to softly stop it, to avoid any software problems and any data lost at switch off. Figure 11

Figure 11



4.5 Communication

The communication between the Data Acquisition System (DAS-PC) and EasyCOD can be done through the serial interface RS 232 or TCP/IP. In order to receive the data through the interface RS 232, following configuration of the interface must be set in both devices (EasyCOD and DAQ) to the same value.

For example:

Baud rate: 19200

Data bits: 8

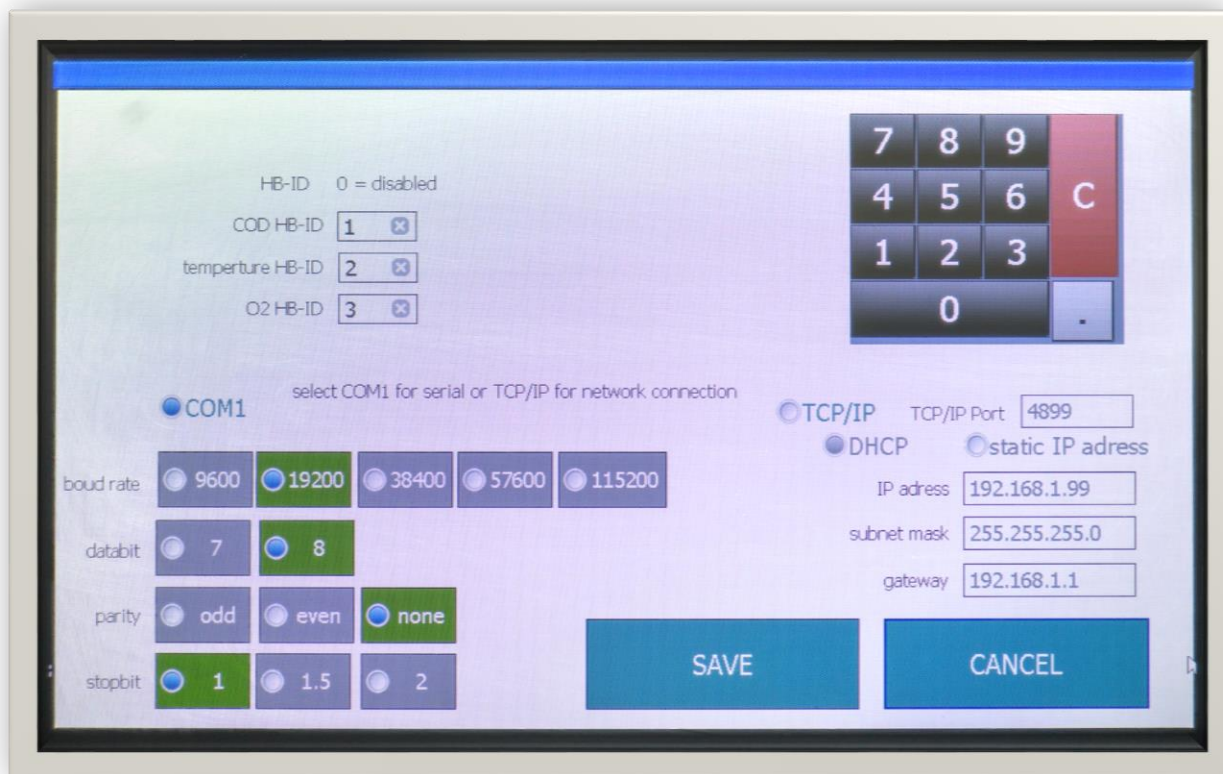
Parity: none

Stop bit: 1

Protocol: HB/Protocol

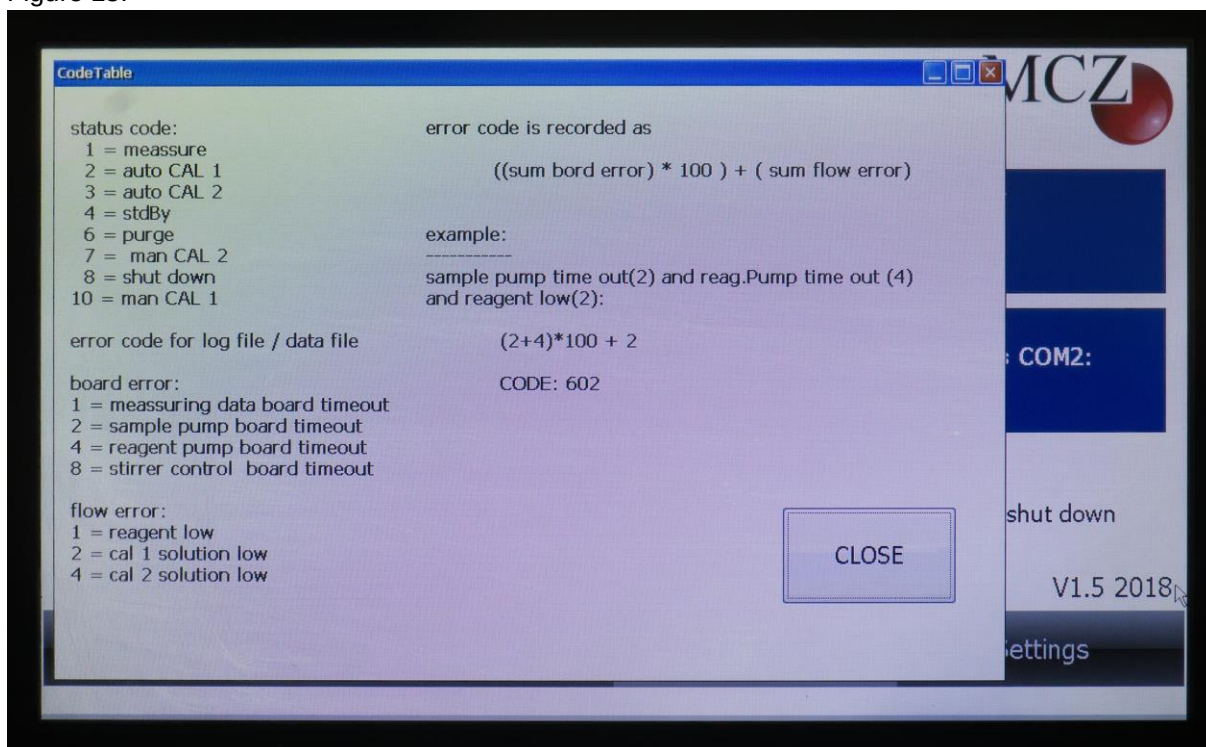
EasyCOD is using HB protocol and the HB ID is already pre set by manufacturer in HKT list. HB ID can be manually set but it is **“very important to set HB ID in both device as same value”**. This settings can be founded in Settings menu. Figure 12

Figure 12



HB status and error codes can be founded in Service/status error code table. Figure 13

Figure 13.

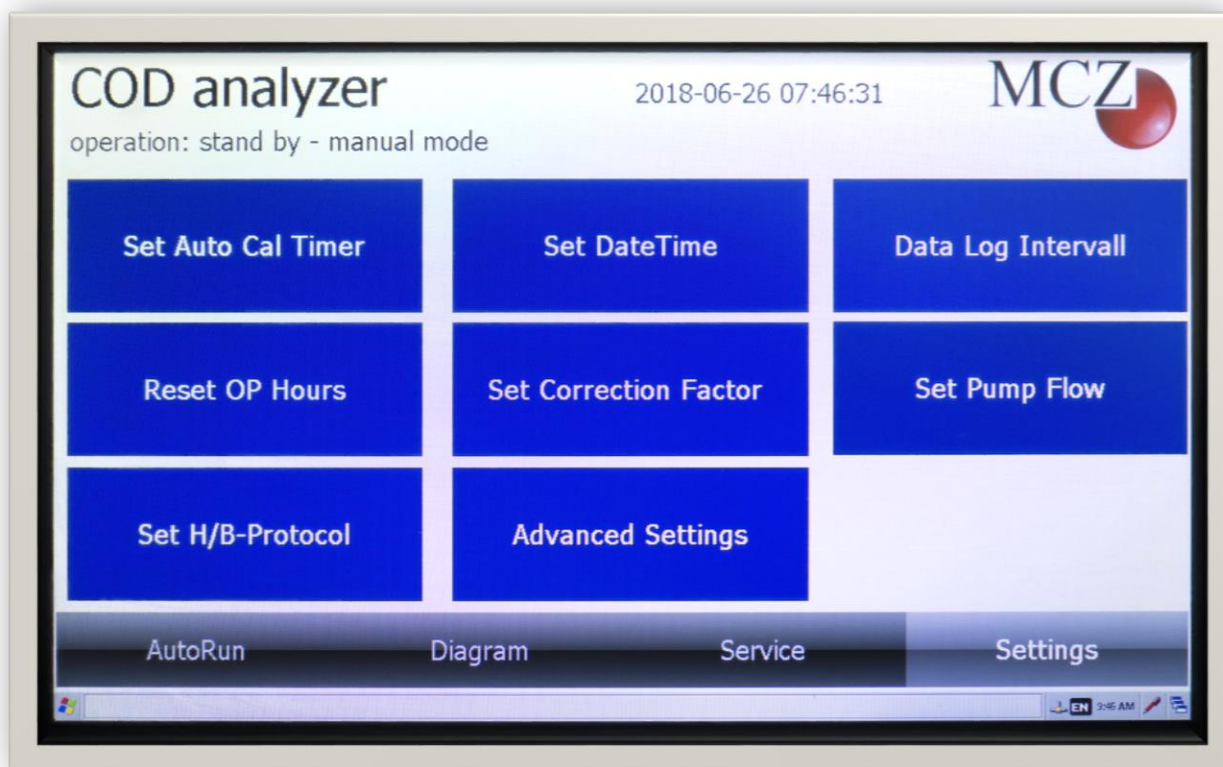


Settings.

In menu window Settings we can find: (figure 14)

- Set auto cal timer.
- Reset OP hours
- Set HB protocol
- Set date and time
- Set correction factor
- Advanced settings
- Data log interval
- Set pump flow

Figure 14

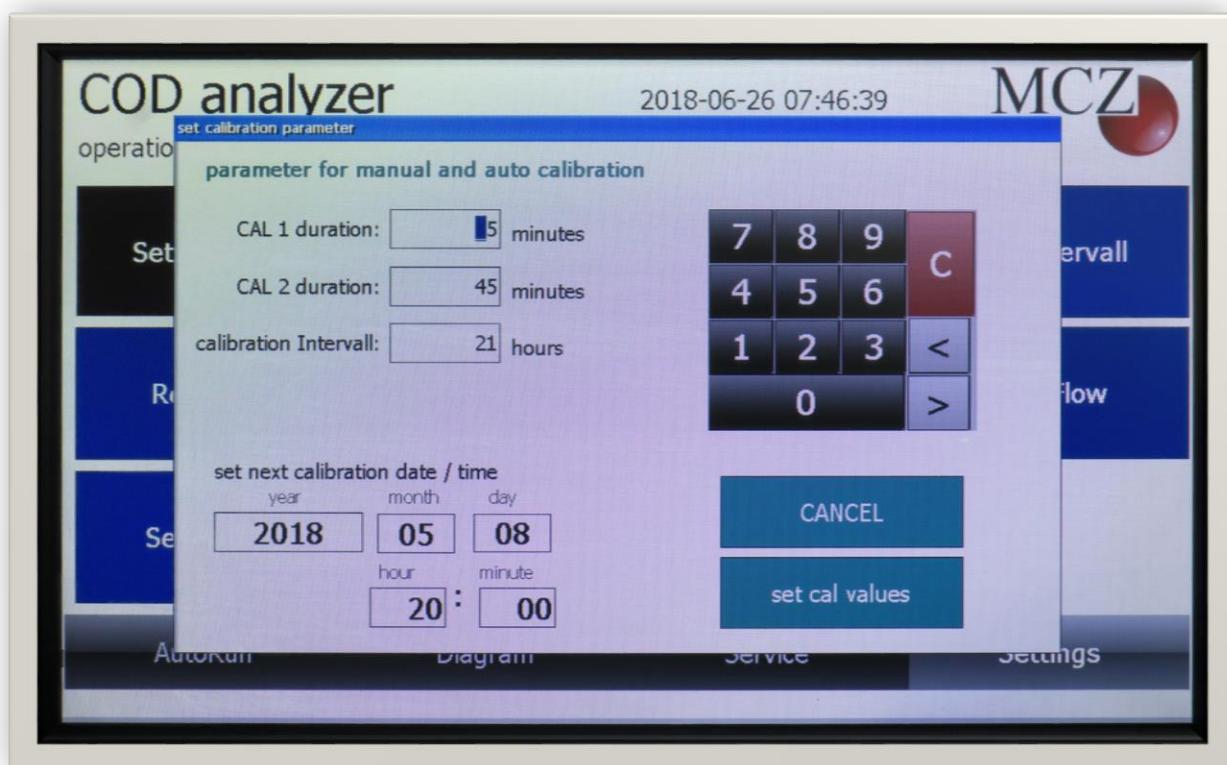


In "Set auto cal timer" menu we can make settings for manual and Auto calibration. Default settings are:

- CAL 1 duration: 5 min
- CAL 2 duration: 60 min
- Calibration interval: 21 hours

We can also set date and time for all next calibrations. MCZ do not recommend to set up CAL 2 duration less then 50 min. Figure 14

Figure 14



With “Reset OP (operation) hours we can, for example, when we change UV lamp or do year maintenance reset OP hours and follow how long the UV lamp will work and/or when is next maintenance.

We can set time and date in settings/Set Date Time menu.

Set Correction factor is by default set to one(1) and it is constant.

Advance settings (figure 15) is protected by **password (1234)** due to if any changes made in this menu will effect work of EasyCOD. It is extremely important before user make any changes in Advanced Settings to backup old settings.

Cal factor will be set up automatically after calibration.

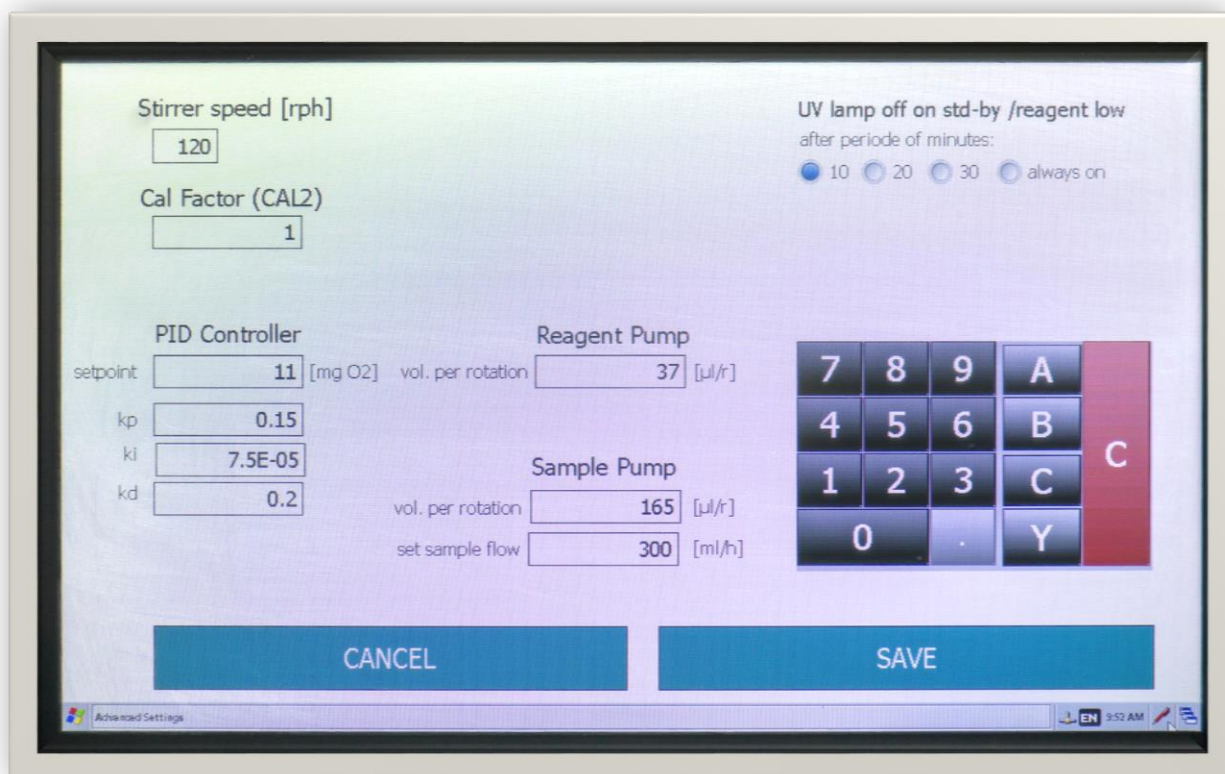
Reagent pump volume per rotation is number that is determine by manufacturer of the hose that we use in reagent pump. We can also recalculate this number in Settings/Set Pump Flow.

Sample pump volume per rotation is number that is determine by manufacturer of the hose that we use in sample pump. We can also recalculate this number in Settings/Set Pump Flow. Figure 16

Set sample flow by default is set to 400 ml/h. We can set sample flow to a larger number that will speed up the system.

PID controller set point is 11 mg/L (see page 8 for explanation).

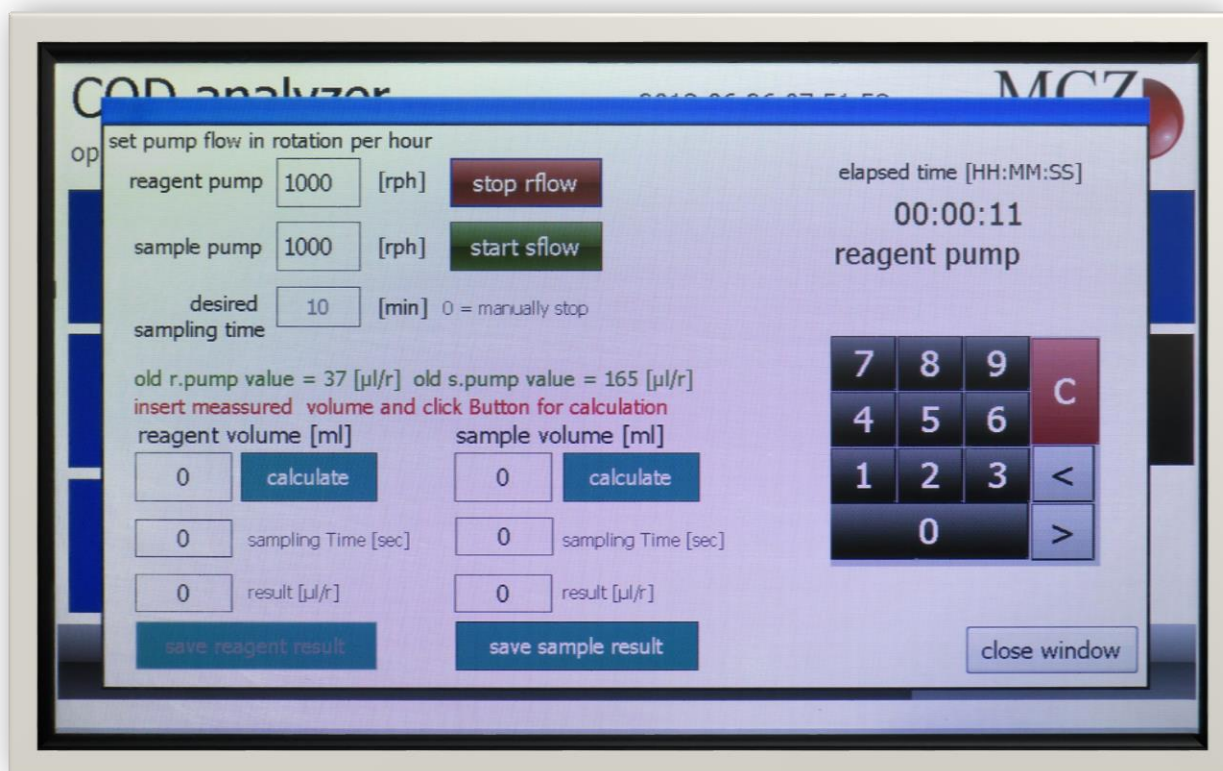
Figure 15



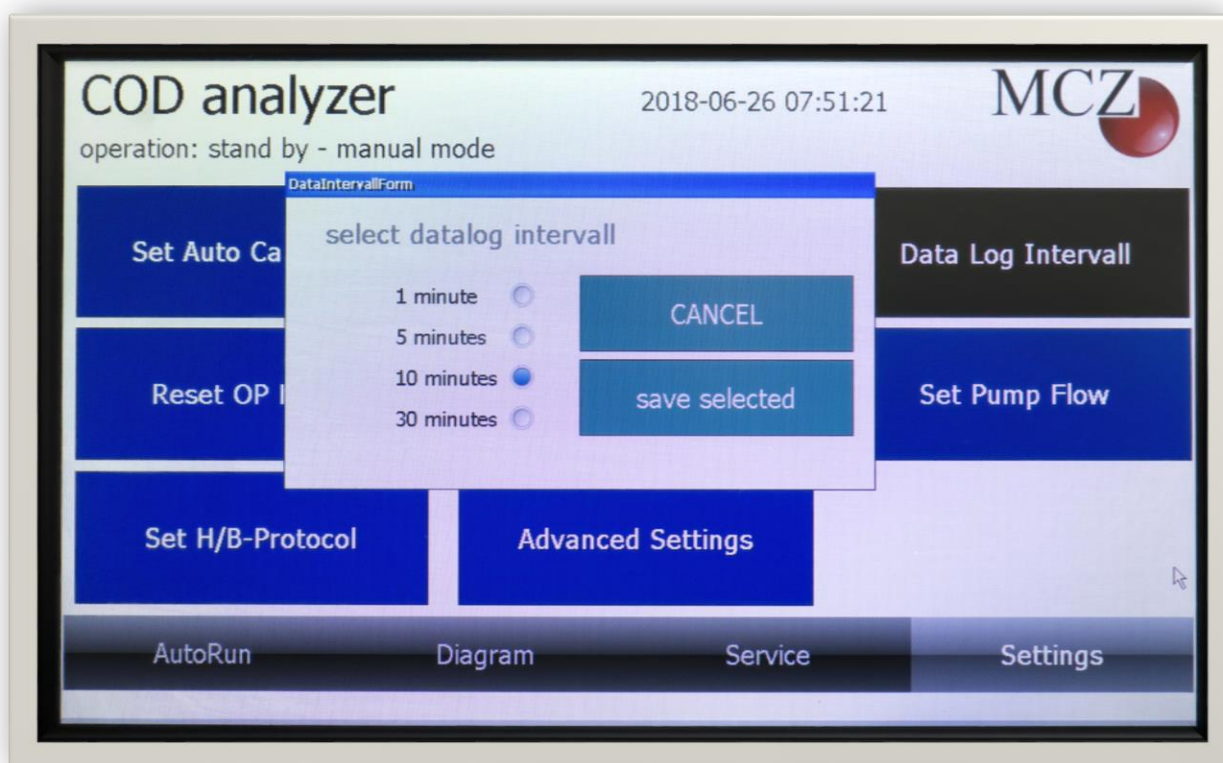
In Advanced settings we can also determine how long our UV lamp will work in std. by mode or when reagent is low.

Note: kp, ki and kd settings are developer settings and that for need to be untouched.

Figure 16



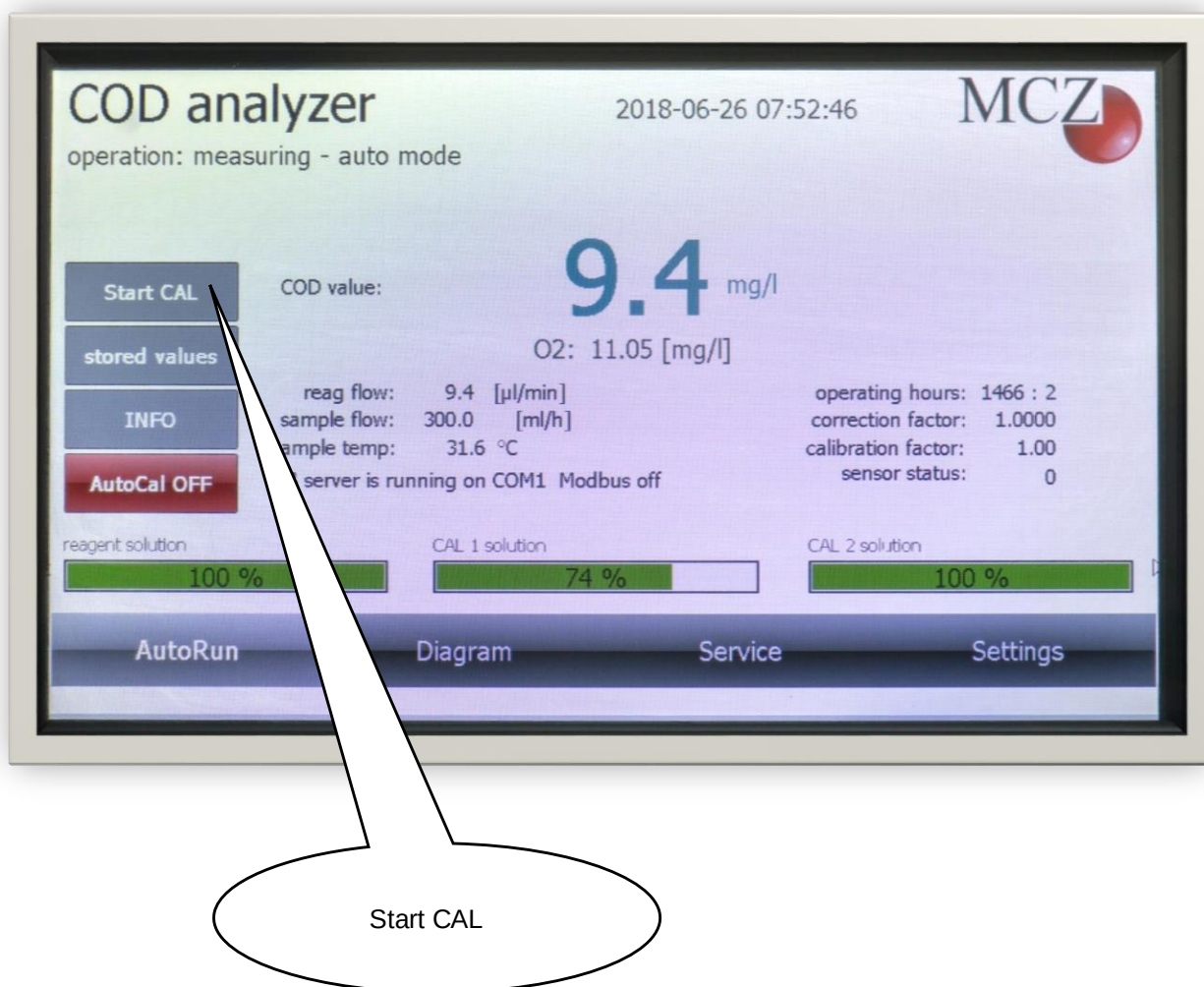
EasyCOD is giving possibility for user to select data logger interval. Figure 17



4.6 Calibration

EasyCOD have automatic calibration. We have to be sure that the CAL 1 and CAL 2 are made by formulas given on page 9. We can also give min/max deviation for O2 value and if the last 10 minutes of calibration are in deviation windows, software will treat it as good calibration and save results. If the last 10 minutes of calibration is not in the deviation window, calibration will not be saved and last calibration will still be actual. On main windows we can start calibration and wait for results. Figure 18

Figure 18



5. Maintenance

For the most part, the EasyCOD analyzer is maintenance free. However it is recommended that a minimal number of simple procedures be performed regularly to ensure that the EasyCOD continues to operate accurately and reliably over its lifetime. In general, the exterior can be wiped down with a lightly damp cloth; avoid spraying anything directly onto any part of the analyzer.

Table 1 shows a typical maintenance schedule for the EasyCOD. Please note that in certain environments some maintenance procedures may need to be performed more often than shown. The following table can be used as a basis.



WARNING – ELECTRICAL SHOCK HAZARD

Disconnect power before performing any of the following operations.



CAUTION–QUALIFIED-PERSONNEL

These maintenance procedures must be performed by qualified technicians only.

Maintenance operation table

Interval	Maintenance Operation	Calibration
As required	-Cleaning the filter -Cleaning the oxygen probe vessel	No
As required 1-4 Weeks	-Depending on COD value, refill reagent -Depending on the calibration cycle, refill calibration solution	No
As required 3 months	-Exchange hoses of the peristaltic pumps -Renew membrane head of the DO probe and calibrate DO probe	Yes
As required 3-6 months	-Clean UV reactor and UV lamp -Exchange the filter	Yes No
Yearly	-Change UV lamp	Yes

Exchange hoses of the peristaltic pump

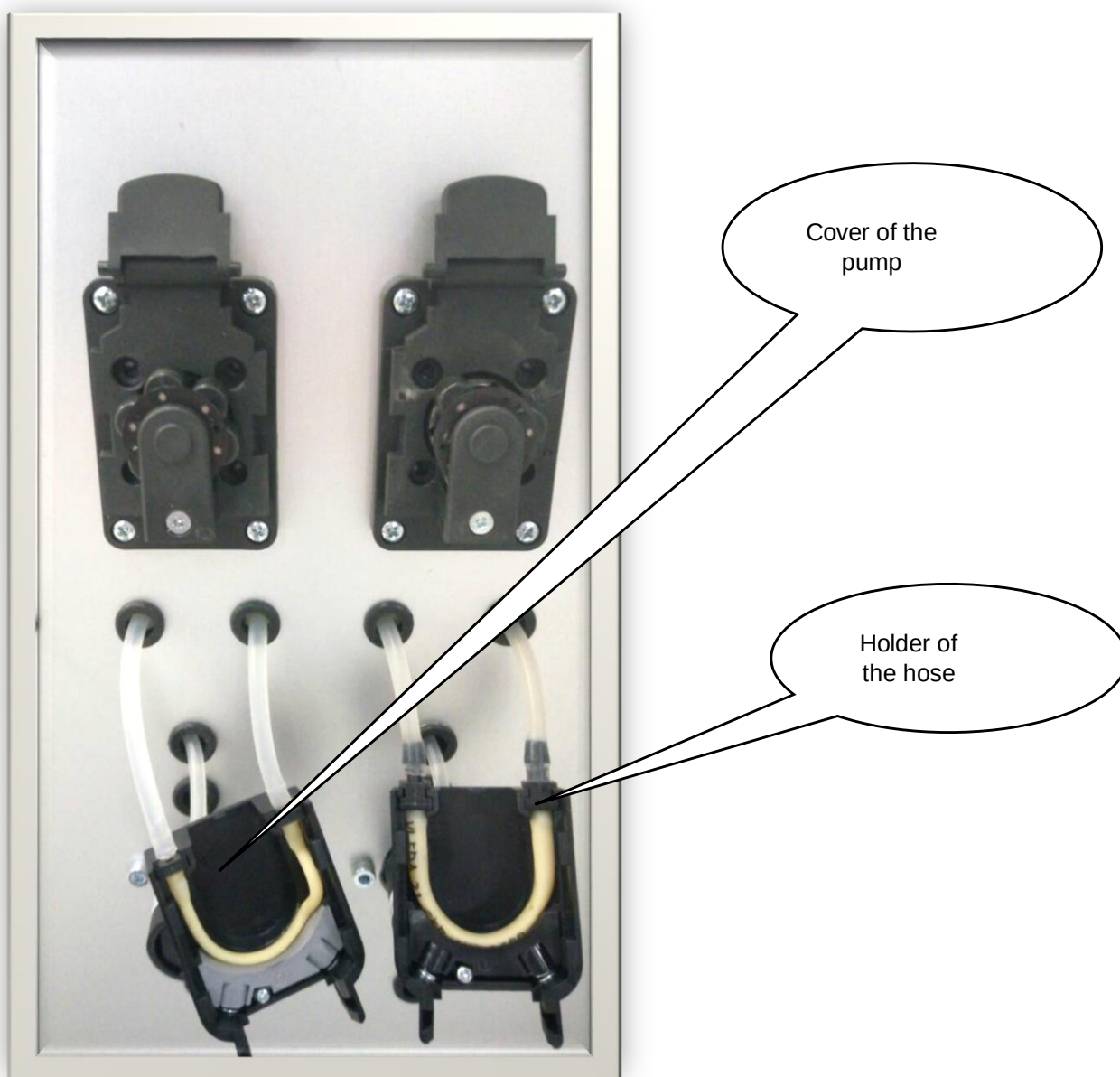
During this procedure analyzer need to go in stand by mode (service or settings), open the cover of the pump and remove from cover the holder of the hose.

Remove old hose and install new one. Put back holder of the hose back to pump cover and put cover back on pump.

Press Settings/Set Pump Flow and set pump on max speed (1800 rpm) for couple of minutes for test.

After procedure is done go back to AutoRun.

Figure 19

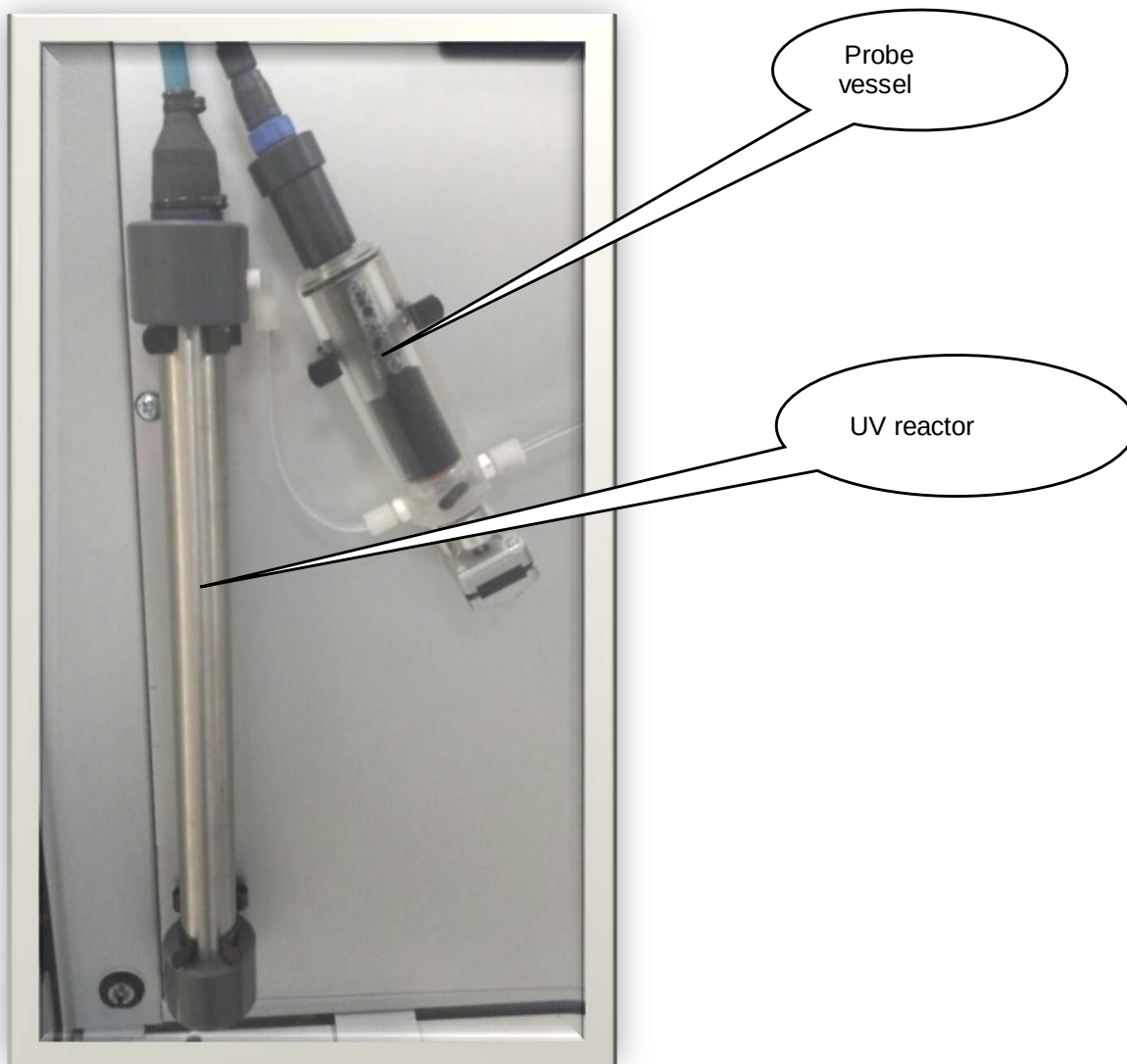


Cleaning the oxygen probe vessel

Depending on the water we are measuring, sometime we need to clean probe vessel. This procedure is quit simple and quit fast.

Analyzer need to go in stand by mode and the remove DO probe from probe vessel. Open two hose on the side of probe vessel and take it out from holders. Wash it in clean water and put back to place.

Figure 20

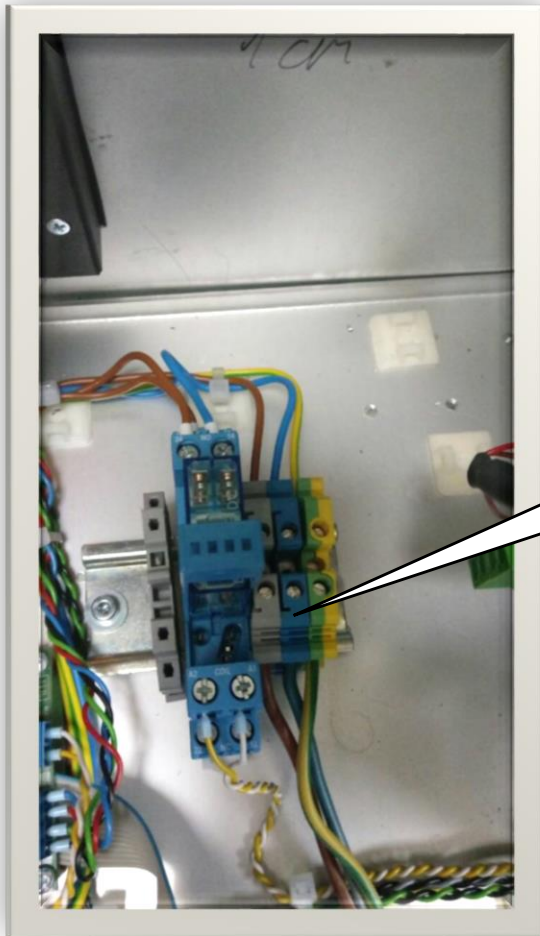


Cleaning the UV reactor

This action also require analyzer to go to stand by. Careful remove the UV lamp from UV reactor (pull up 90 degrees) and remove hose from reactor. Take of reactor from holder and clean it with water. Same way put all back and wait couple of hours then recalibrate.

UV lamp exchange

For this action analyzer require to be power off. Remove 4 screws from electronic cover and find UV lamp input wires showed in figure 21 and 22. Remove wires from clamps then remove lamp from UV reactor. New lamp comes with long wire so we need to cut wire and prepare it for installation. After preparing new UV lamp install it back and wait for 2-3 hours then recalibrate analyzer.



UV lamp clamps

Exchange the filter

Changing the filter is very easy and can be done in couple of minutes. Analyzer need to be in stand by mode. Remove the filter cap and remove the filter. Install new filter and close the cap. Figure 23

Figure 23

